

Astra evidence inventory - kelvin-hk

account 709569057388 · us-east-1 · generated 2026-06-25 12:20

Security

A001

IAM role titanflow-astra-worker-role exists

• titanflow-astra-worker-role

arn=

note - Naming - be lenient: accept any role name that follows the titanflow-astra-* convention. The grader keys on titanflow-astra-worker-role, but a convention-following name is equally acceptable - do not penalise it.

aws

OK

A002

Worker role trust policy IRSA-bound to astra-worker-sa

• titanflow-astra-worker-role

role missing

aws

OK

A003

Worker policy: dynamodb:PutItem scoped to worker-registry only

• titanflow-astra-worker-role

action=PutItem

resource=titanflow-astra-worker-registry

inline+managed_docs=0

granted_by=None

fetch_errors=list-role-policies(titanflow-astra-worker-role): aws: [ERROR]: An error occurred (NoSuchEntity)

when calling the ListRolePolicies operation: The role with name titanflow-astra-worker-role cannot be found., list-attached-role-policies(titanflow-astra-

worker-role): aws: [ERROR]: An error occurred (NoSuchEntity) when calling the

ListAttachedRolePolicies operation: The role with name titanflow-astra-worker-role cannot be found.

no scoped grant found

aws

OK

A004

Worker policy: GetSecretValue scoped to floor-secret only

• titanflow-astra-worker-role

action=GetSecretValue

resource=titanflow-astra-floor-secret

inline+managed_docs=0

granted_by=None

fetch_errors=list-role-policies(titanflow-astra-worker-role): aws: [ERROR]: An error occurred (NoSuchEntity)

when calling the ListRolePolicies operation: The role with name titanflow-astra-worker-role cannot be found., list-attached-role-policies(titanflow-astra-

worker-role): aws: [ERROR]: An error occurred (NoSuchEntity) when calling the

ListAttachedRolePolicies operation: The role with name titanflow-astra-worker-role cannot be found.

no scoped grant found

aws

OK

note - Naming - be lenient: accept any names that follow the titanflow-astra-* convention for the role/secret references; do not penalise a convention-following name that differs from the exact string.

A005

IAM role titanflow-astra-router-role exists

• titanflow-astra-router-role

arn=

aws

OK

A006

Router role trust policy IRSA-bound
to astra-router-sa

- titanflow-astra-router-role
role missing

aws

OK

A007

Router policy: DynamoDB *Item
scoped to trades table only

- titanflow-astra-router-role
action=Item
resource=titanflow-astra-trades
inline+managed_docs=0
granted_by=None
fetch_errors=list-role-policies(titanflow-astra-router-
role): aws: [ERROR]: An error occurred (NoSuchEntity)
when calling the ListRolePolicies operation: The role
with name titanflow-astra-router-role cannot be found.,
list-attached-role-policies(titanflow-astra-router-role):
aws: [ERROR]: An error occurred (NoSuchEntity) when
calling the ListAttachedRolePolicies operation: The role
with name titanflow-astra-router-role cannot be found.
per_action= GetItem=False
 PutItem=False
 BatchWriteItem=False
service_wildcard_grant=False
missing: GetItem,PutItem,BatchWriteItem

aws

OK

A008

Neither workload role has a
wildcard Action/Resource

- titanflow-astra-worker-role
inline+managed_docs=0
wildcards=[]
fetch_errors=list-role-policies(titanflow-astra-worker-
role): aws: [ERROR]: An error occurred (NoSuchEntity)
when calling the ListRolePolicies operation: The role
with name titanflow-astra-worker-role cannot be
found., list-attached-role-policies(titanflow-astra-
worker-role): aws: [ERROR]: An error occurred
(NoSuchEntity) when calling the
ListAttachedRolePolicies operation: The role with name
titanflow-astra-worker-role cannot be found.
no wildcard

aws

OK

- titanflow-astra-router-role
inline+managed_docs=0
wildcards=[]
fetch_errors=list-role-policies(titanflow-astra-router-
role): aws: [ERROR]: An error occurred (NoSuchEntity)
when calling the ListRolePolicies operation: The role
with name titanflow-astra-router-role cannot be found.,
list-attached-role-policies(titanflow-astra-router-role):
aws: [ERROR]: An error occurred (NoSuchEntity) when
calling the ListAttachedRolePolicies operation: The role
with name titanflow-astra-router-role cannot be found.
no wildcard

A009

ServiceAccount astra-worker-sa
annotated with worker role ARN

- astra-worker-sa
role-arn=
namespace=None
duplicate_sa_names=0

note - Worker IRSA: role must carry DynamoDB (PutItem
on worker-registry) + Secrets Manager (GetSecretValue
on floor-secret); ServiceAccount astra-worker-sa must be
annotated with the worker-role ARN.

k8s

OK

A010

ServiceAccount astra-router-sa
annotated with router role ARN

- astra-router-sa
role-arn=
namespace=None
duplicate_sa_names=0

k8s

OK

note - Router IRSA: role must carry DynamoDB only
(GetItem/PutItem/BatchWriteItem on trades, no Secrets
Manager); ServiceAccount astra-router-sa must be
annotated with the router-role ARN.

A011

Redis security group: inbound 6379
from cluster only (not 0.0.0.0/0)

(no matching resources found)

aws

OK

A012

WAFv2 REGIONAL WebACL
titanflow-astra-waf present

- astra_rule
id=4b5d696b-f6af-4728-9b78-381c687e6554

aws

OK

A013

WebACL includes
AWSManagedRulesCommonRuleSet

- astra_rule
managed_groups=AWSManagedRulesAntiDDoSRuleSet,
AWSManagedRulesAmazonIpReputationList,
AWSManagedRulesAnonymousIpList,
AWSManagedRulesCommonRuleSet,
AWSManagedRulesBotControlRuleSet

aws

OK

A014

WebACL associated to the API
Gateway prod stage

- astra_rule
associated=arn:aws:apigateway:us-east-1::/
restapis/6bpqn7x3ig/stages/prod
expected_prod_suffix=None
matched_prod_stage_arn=None
REST API id unresolved

aws

OK

A015

VPC Flow Logs to a CloudWatch
Logs group

- fl-0df882d206db5309a
resource=vpc-0693f15ca01868cc7
dest=cloud-watch-logs
log_group=/aws/vpc/flowlogs/astra
on_dedicated_vpc=True
target_vpc=vpc-0693f15ca01868cc7
status=ACTIVE
deliver_error=None

aws

OK

A016

Both ECR repositories have
scanOnPush = true

- astra-worker
scanOnPush=False
registry_scan_type=BASIC
registry_covered=False
- astra-router
scanOnPush=False
registry_scan_type=BASIC
registry_covered=False

aws

OK

A017

No Lambda outside the clearing
workflow (prefix + referenced)

- (none)
note=zero Lambda - trading path is Lambda-free

aws

OK

A018

Clearing verify role: GetSecretValue

(no matching resources found)

aws

OK

scoped to clearing-secret only

A019	(no matching resources found)	aws
Clearing state-machine role is least-privilege (no wildcards)		OK

Performance Efficiency

A020	• astra-workers-headless clusterIP=None ports=[] selector=None	k8s
Headless Service astra-workers-headless (clusterIP None)		OK

A021	• ROUTING_MODE value=optimized	k8s
Router env ROUTING_MODE = optimized		OK

A022	• titanflow-astra-redis-cache kind=serverless engine=redis status=available	aws
ElastiCache cache titanflow-astra-redis exists (Serverless or node)		OK

A023	• titanflow-astra-redis-cache subnets= • subnet=subnet-05e44c1d0666f4c98 vpc=vpc-0693f15ca01868cc7 private=True • subnet=subnet-0a175b1e2d49ade54 vpc=vpc-0693f15ca01868cc7 private=True dedicated_vpc=vpc-0693f15ca01868cc7 ln_dedicated_vpc=True all_private=True	aws
Cache in the dedicated VPC's private subnets		OK

A024	• worker:scores ZSET routing_match=0 routing_near=0 routing_stale=0 routing_samples=0 note=routing_match>0 implies the router read a populated worker:scores zset and picked the fastest floor manual - No routing outcomes sampled yet - cannot infer the zset is populated; verify via the router pod / routing-match rate.	live
Cache reachable; router connected (worker:scores populated)		MANUAL

A025	• REDIS_HOST value=titanflow-astra-redis-cache-dlbi2p.serverless.us-east-1.cache.amazonaws.com valueFrom=False • REDIS_PORT value=6379 valueFrom=False	k8s
Router env REDIS_HOST/REDIS_PORT set correctly		OK

A026	Routing-match rate $\geq$ 90% (top tier)	• routing match routing_samples=0 manual - No routing outcomes sampled yet.	live MANUAL
A027	Zero routing_stale outcomes (recent window)	• routing_stale stale_count=0 routing_samples=0 note=counted over the recent logged-request window, not strictly the final hour; scheme intent is 'over the final hour' manual - No routing outcomes sampled yet - cannot assert zero routing_stale (insufficient evidence, not a pass).	live MANUAL
A028	p99_ms < 500 (tight tier)	• p99 < 500ms p99_ms=24.6 note=engine rolling-sample p99; confirm it reflects the run-wide p99 (scheme intent is 'over the run')	live OK
A029	HPA astra-workers-hpa targets the astra-workers Deployment	(no matching resources found)	k8s OK
A030	astra-workers-hpa is bounded (max > min, within ~15-floor capacity)	(no matching resources found)	k8s OK
A031	astra-workers-hpa scales on a CPU-utilization metric	(no matching resources found)	k8s OK
A032	HPA astra-router-hpa targets the astra-router Deployment	(no matching resources found)	k8s OK
A033	astra-router-hpa is bounded (max > min)	(no matching resources found)	k8s OK
A034	astra-router-hpa scales on a CPU-utilization metric	(no matching resources found)	k8s OK
A035	Worker container sets a CPU request (so the HPA can measure)	• astra-workers cpu_requests=[]	k8s OK
A036	Routing-match rate $\geq$ 60%	• routing match routing_samples=0 manual - No routing outcomes sampled yet.	live MANUAL
A037	Routing-match rate $\geq$ 80% (higher tier)	• routing match routing_samples=0 manual - No routing outcomes sampled yet.	live MANUAL

A038	p99_ms < 1000	- p99 < 1000ms - p99_ms=24.6 - note=engine rolling-sample p99; confirm it reflects the run-wide p99 (scheme intent is 'over the run')	live OK
A039	High-value clearing live (>=1 settlement cleared)	- clearing live - milestones=tag_set, endpoint_discovered, first_success	live OK
Reliability			
A040	Deployment astra-workers exists	- astra-workers - readyReplicas=0	k8s OK
A041	astra-workers pods use astra-worker:1.0.0 + astra-worker-sa	(no matching resources found)	k8s OK
A042	astra-workers pods are Running	- astra-workers - running=0 - total=0 - desired_replicas=None 0 running	k8s OK
A043	worker-registry shows fresh rows (recent ts_ms)	- worker-registry - rows=4 - fresh_rows=4 - rows_ts_in_seconds=0 - rows_ts_unparseable=0 - fresh heartbeats present	aws OK
A044	WORKER_ID not hardcoded (distinct ids in registry)	- distinct worker_ids - distinct_fresh=4 - distinct_total=4 - sample=astra-worker-6f946d6984-lw8zn, astra-worker-6f946d6984-fc4c6, astra-worker-6f946d6984-xhnrg, astra-worker-6f946d6984-8p5q6 - distinct per replica	aws OK
A045	Deployment astra-router exists	- astra-router - readyReplicas=2	k8s OK
A046	astra-router pods use astra-router:1.0.0 + astra-router-sa	(no matching resources found) note - Image tag; any valid tag is acceptable - the pods need not use :1.0.0 specifically. The grader matches :1.0.0, so accept the correct router image under any tag by grader judgement.	k8s OK
A047	astra-router runs >= 2 replicas	- astra-router - desired_replicas=2 - ready_replicas=2 2+ ready	k8s OK

A048	DynamoDB titanflow-astra-trades is ACTIVE	titanflow-astra-trades status=ACTIVE	aws OK
A049	trades partition key = trade_id (String)	trade_id key= trade_id=HASH type=S	aws OK
A050	Internet-facing ALB titanflow-astra-alb via the Ingress	- k8s-default-astrarou-a1a647623b scheme=internal type=application state=active ingress_provenance_tags=True not a matching ALB - k8s-astrarou-astrarou-a2a48048be scheme=internet-facing type=application state=active ingress_provenance_tags=True not a matching ALB	aws OK
A051	WebACL rate-based rule caps a single IP	astra_rule rate_limit=1000 aggregate_key=IP evaluation_window_sec=300 rate rule present but limit=1000 agg=IP window=300	aws OK
A052	Token & settle/quarantine separate states with Retry/Catch	(no matching resources found) manual - needs grader judgement	aws MANUAL
A053	GET /workers end-to-end returns a non-empty floor list	GET /workers status=200 count=4 url=https://6bpqn7x3ig.execute-api.us-east-1.amazonaws.com/prod/workers ssm_api_endpoint=https://6bpqn7x3ig.execute-api.us-east-1.amazonaws.com/prod via_api_gateway=True non-empty floor list via API Gateway	aws OK
A054	AWS Backup vault titanflow-astra-vault exists	titanflow-astra-vault vaults=[]	aws OK
A055	Backup plan selects trades with a defined retention	(no matching resources found)	aws OK
A056	Failure rate fail/sent <= 5%	fail/sent <= 5% fail_pct=40.77 sent=601573 fail=245235	live OK

A057

Read-your-writes holds (ryw-failure
< 1%)

- ryw
matched_outcome_keys=[]
outcome_keys_seen=no_cache_ok, ok

manual - No submissions sampled.

live

MANUAL

Operational Excellence

A058

astra-worker:1.0.0 image present in
ECR

- astra-worker
tags=1.0.0

aws

OK

A059

astra-router:1.0.0 image present in
ECR

- astra-router
tags=1.0.0

aws

OK

A060

Both ECR repos imageTagMutability
= IMMUTABLE

- astra-worker
imageTagMutability=IMMUTABLE
- astra-router
imageTagMutability=IMMUTABLE

aws

OK

A061

Worker env sets
REGISTRY_TABLE_NAME +
FLOOR_SECRET_NAME

(no matching resources found)

k8s

OK

A062

Router env sets
WORKERS_SERVICE_DNS +
TRADES_TABLE_NAME

- WORKERS_SERVICE_DNS
value=astra-worker.astra-worker.svc.cluster.local
valueFrom=False
- TRADES_TABLE_NAME
value=titanflow-astra-trades
valueFrom=False

k8s

OK

A063

Ingress astra-router-ingress
materialised by the LB controller

- astra-router-ingress
materialised=True
routes_to_astra_router_svc_8080=False
paths=
 - path=/submit
service=astra-router
port=80
 - path=/workers
service=astra-router
port=80
 - path=/trades
service=astra-router
port=80

k8s

OK

A064

REST API titanflow-astra-api with a
prod stage

- titanflow-astra-api
api_id=None
prod_stage=False
proxy_resource=False
any_method=False
http_proxy_integration=False
integration_url=

aws

OK

A065

SSM /titanflow/api_endpoint = valid
prod invoke URL

- /titanflow/api_endpoint
value=https://6bpqn7x3ig.execute-api.us-east-1.amazonaws.com/prod
has_execute_api=True
has_prod_stage=True
rest_api_id=None
api_id_matches=None

aws

OK

A066

Container Insights publishing with
recent streams

- container-insights
groups=[]
recent_event_epoch_ms=None
stream_age_s=None
fresh_within_10min=False
node_role_has_CloudWatchAgentServerPolicy=False
addon_irsa_role=False
addon_pod_identity_assoc=False

aws

OK

note - Permission mechanism: the scheme text names 'node role CloudWatchAgentServerPolicy or addon IRSA role', but a Pod Identity Association on the addon is equally valid. Accept any of the three (node role / IRSA / Pod Identity) - the verdict gates on fresh log streams, not the mechanism (see addon_irsa_role / addon_pod_identity_assoc).

A067

Alert pipeline: titanflow-astra-alerts
-> alert-sink SQS

- titanflow-astra-alerts
topic=False
sqs_subscribed=False

aws

OK

A068

CloudWatch alarms cover ALB/pod/
DDB/healthy-host

- ALB 5XX
matching_alarms=[]
actioned_to_alerts=[]
no alarm for this metric
- worker pod CPU
matching_alarms=[]
actioned_to_alerts=[]
no alarm for this metric
- DynamoDB throttling
matching_alarms=[]
actioned_to_alerts=[]
no alarm for this metric
- ALB healthy-host
matching_alarms=[]
actioned_to_alerts=[]
no alarm for this metric

aws

MANUAL

manual - needs grader judgement

A069

Express state machine titanflow-
astra-clearing + sync route

- titanflow-astra-clearing
type=None
is_express=False
clearing_route_present=False
synchronous_start=False
post_integration_uri=

aws

OK

A070

Clearing definition: verify -> Choice
-> settle/quarantine + SNS

(no matching resources found)

manual - needs grader judgement

aws

MANUAL

A071

X-Ray tracing enabled on the prod stage

• prod stage
tracingEnabled=False

aws

OK

A072

Dashboard titanflow-astra-dashboard exists

• titanflow-astra-dashboard
dashboards=astra
widget_categories= ALB=False
router/floor CPU=False
trades capacity=False
cache=False

aws

MANUAL

manual - needs grader judgement

A073

ALB access logs delivered to the S3 logs bucket

• k8s-default-astrarou-a1a647623b
access_logs_enabled=False
bucket=
bucket_name_matches_logs=False
bucket_policy_grants_log_delivery=None
bucket_policy_principals=[]
access logs disabled

• k8s-astrarou-astrarou-a2a48048be
access_logs_enabled=False
bucket=
bucket_name_matches_logs=False
bucket_policy_grants_log_delivery=None
bucket_policy_principals=[]
access logs disabled

aws

OK

A074

S3 access-log bucket titanflow-astra-logs-NNNN exists

(no matching resources found)

aws

OK

Work Organisation and Management

A075

All competitor resources carry CompetitorId tag

• stack/eksctl-titanflow-astra-cluster-addon-iamserviceaccount-astra-router-astra-router-role/3c03ce20-7046-11f1-8f86-12ca0f276931
arn=arn:aws:cloudformation:us-east-1:709569057388:stack/eksctl-titanflow-astra-cluster-addon-iamserviceaccount-astra-router-astra-router-role/3c03ce20-7046-11f1-8f86-12ca0f276931
CompetitorId=None
expected=kelvin-hk
no CompetitorId tag

• titanflow-astra-redis-cache
arn=arn:aws:elasticache:us-east-1:709569057388:serverlesscache:titanflow-astra-redis-cache
CompetitorId=None
expected=kelvin-hk
no CompetitorId tag

• stack/eksctl-titanflow-astra-cluster-addon-

aws

OK

iamserviceaccount-astra-worker-astra-worker-
role/fc292ad0-7035-11f1-bfcb-0ead9eac1909
arn=arn:aws:cloudformation:us-
east-1:709569057388:stack/eksctl-titanflow-astra-
cluster-addon-iamserviceaccount-astra-worker-astra-
worker-role/fc292ad0-7035-11f1-bfcb-0ead9eac1909
CompetitorId=None
expected=kelvin-hk
no CompetitorId tag

A076	Resources follow titanflow-astra-* naming	• naming titanflow-astra arns=43 total=107 manual - Evidence: resource ARNs for naming- convention review (no CompetitorId-tagged resources found).	aws MANUAL
A077	No partial/orphaned resources remain	• astra ALBs items=k8s-default-astrarou-a1a647623b, k8s- astrarou-astrarou-a2a48048be count=2 multiple astra load balancers manual - needs grader judgement	aws MANUAL
A078	Live placement: rank = 1 at session end	• rank <= 1 rank=4 of=8 score=-34211 top_score=1618547 tied_at_top=False	live OK
A079	Live placement: rank <= 2	• rank <= 2 rank=4 of=8 score=-34211 top_score=1618547 tied_at_top=False	live OK
A080	Live placement: rank <= 3	• rank <= 3 rank=4 of=8 score=-34211 top_score=1618547 tied_at_top=False	live OK
A081	Engine.sent >= 20,000	• sent >= 20000 sent=601573	live OK
A082	Engine.sent >= 40,000	• sent >= 40000 sent=601573	live OK
A083	Clearing resources follow the naming convention	• state machine name=titanflow-astra-clearing	aws OK

A084

Both images pushed to ECR in the first hour

- **astra-worker**
earliest_imagePushedAt_epoch=1782344820.038
image_count=1
tags=1.0.0
timestamp evidence - compare to session start

- **astra-router**
earliest_imagePushedAt_epoch=1782345413.783
image_count=1
tags=1.0.0
timestamp evidence - compare to session start

manual - Evidence: earliest image push time per repo (grader compares to session start +1h).

MANUAL

Cost Optimisation

A085

trades uses PAY_PER_REQUEST billing

- **titanflow-astra-trades**
billing=PAY_PER_REQUEST

aws

OK

A086

ElastiCache uses the Valkey engine

- **titanflow-astra-redis-cache**
engine=redis

aws

OK

A087

ElastiCache ranking store is right-sized

- **titanflow-astra-redis-cache**
kind=serverless
note=right-sized by design
manual - needs grader judgement

aws

MANUAL

A088

Every clearing Lambda log group has bounded retention

(no matching resources found)

aws

OK

A089

S3 logs lifecycle: transition to ONEZONE_IA at day 30

(no matching resources found)

aws

OK

A090

Worker/router pods set sized CPU/memory requests

- **astra-workers**
requests=[]
missing cpu/memory request
- **astra-router**
requests=
 - cpu=250m
memory=200Mi
 cpu+memory requests set
manual - needs grader judgement

k8s

MANUAL

A091

HPA minReplicas set to a lean baseline

(no matching resources found)
manual - needs grader judgement

k8s

MANUAL

A092

Backup plan retention = 7 days

(no matching resources found)

aws

OK

A093

Clearing Lambdas right-sized
(memory/timeout)

(no matching resources found)
.....
manual - needs grader judgement

aws

MANUAL

Sustainability

A094

trades has TTL enabled on
expires_at

• titanflow-astra-trades
ttl= TimeToLiveStatus=ENABLED
AttributeName=expires_at

aws

OK

A095

S3 logs lifecycle: expiry at day 90

(no matching resources found)

aws

OK

A096

Both ECR repos have an image-
expiry lifecycle policy

• astra-worker
has_policy=False
rules=[]
no lifecycle policy

aws

OK

• astra-router
has_policy=False
rules=[]
no lifecycle policy

A097

Container Insights application log
group bounded retention

(no matching resources found)

aws

OK

A098

VPC Flow Logs log group bounded
retention

(no matching resources found)

aws

OK

A099

Workers HPA scales down to
minimum at low load

(no matching resources found)

k8s

OK

A100

TTL actually reclaiming (no rows
past expires_at)

• expired rows present
rows_past_expiry=0
scanned_count=19057
pages_scanned=1
no rows past expiry

aws

OK